ATTICUS ROTHSCHILD

ar311@rice.edu | +1 (917)-502-9209

EDUCATION

Rice University | **Houston, TX** *Aug 2025 - Present*
B.S. Mechanical Engineering (expected 2029)
- Rice Racing (Formula SAE) - Subteam Lead, Powertrain

Columbia University | **New York, NY** *Aug 2022 - Dec 2022*
Completed Coursework
- Data Structures and Algorithms in Java
- Introduction to Electrical Engineering

EXPERIENCE

Rice University - EMERGE Lab | **Houston, TX** *Jun 2026 - Present*
Undergraduate Researcher
- Designed a precision micromanipulation system and light-tight enclosure in CAD for repeatable positioning and optical characterization of printed MoS2 microsensors.
- Developed an opto-mechanical test setup to measure light transmission and absorption through MoS2 layers for controlled drug-delivery microsensor research.
- Performed cleanroom microfabrication using photolithography, photoresist processing, and thin-film deposition.

Rice University - Kong Research Group | **Houston, TX** *Jun 2026 - Aug 2026*
Undergraduate Researcher
- Supported development of printed microsensor technology for controlled drug-delivery applications as part of a university research team.
- Contributed to development of an insulating layer for printed microsensors to support reliable device isolation and integration.

FIRE1 | **Dublin, Ireland** *May 2026*
Medical Device Engineering Work Experience
- Collaborated with the implantation team to troubleshoot NORM Sensor deployment in the inferior vena cava (IVC), investigating a specialized titanium-alloy frame that folded during catheter launch and prevented full expansion.
- Evaluated alternative catheter-based implantation methods to improve sensor launch and deployment reliability.

Mechanical Engineering Apprentice | **Chatham, NY** *Jun 2024 - 2026*
Apprentice to Lee Gagne, Mechanical Engineer
- Assisted senior engineer in teardown, inspection, and reassembly of combustion engines and automotive systems.
- Performed precision measurement, component refurbishment, and system testing under engineering supervision.

Rapyuta Robotics | **Tokyo, Japan** *Jun 2023 - Dec 2023*
Simulations and Software Integration Intern
- Designed and integrated new features into robot simulation in Unreal Engine, increasing robot running count 300%.
- Conducted real-world and simulated robot demonstrations for prospective clients.

FTC Robotics Team *2022 - 2024*
- FTC Robotics Championships (2023 - 2024)
- FTC Robotics Super Qualifiers (2022 - 2023)

TECHNICAL SKILLS

Software Languages | C++, Java, HTML, Python
Tools & Frameworks | Git, Bash, Docker, ROS, CI/CD
3D Software | Fusion 360, SolidWorks, Unreal Engine, Blender
Manufacturing Tools | Prusa/Ultimaker 3D printers, Epilog laser cutters, mill, and other shop tools

PROJECTS

- Designed and fabricated high-fidelity loudspeakers using CNC-routed MDF enclosures, passive crossover design, and acoustic measurement for performance tuning.
- Amateur radio operator licensed at Amateur Extra level; built and tuned multi-band transceivers.